

Findings

VS and LB — the third observation class

The class that holds MEASUREMENTS. It brings the biggest structural change in the whole course: the raw data must be turned on its side.

By the end of this module you can...

1

Recognise a Findings domain and name the topic variable that makes it one.

2

Transpose wide raw data to tall SDTM the single biggest structural change you will make.

3

Explain --ORRES vs --STRESC vs --STRESN why one result is stored three times, and what each is for.

4

Derive --BLFL correctly the baseline flag that every efficacy and safety analysis depends on.

5

Read a reference range and derive --NRIND from it.

Three classes, three kinds of sentence

INTERVENTIONS

Something was GIVEN

--TRT

topic variable

EX · CM · PR

"Subject 001 was given Drug A, 50 mg, orally, once daily."

EVENTS

Something HAPPENED

--TERM

topic variable

AE · MH · DS

"Subject 001 had a bad headache on 15 March."

FINDINGS

Something was MEASURED

--TESTCD / --TEST

topic variable

VS · LB · EG · PE

"Subject 001's systolic blood pressure was 120 mmHg at Baseline."

Findings is by far the LARGEST class — most SDTM domains, and most rows in a typical submission, are Findings.

THE PATTERN

Same skeleton you already know — one part changes

IDENTIFIERS

Who and which record

STUDYID · DOMAIN · USUBJID · VSSEQ

TOPIC

WHICH TEST was performed

VSTESTCD · VSTEST

RESULT

WHAT THE TEST RETURNED

VSORRES · VSSTRESC · VSSTRESN · VSORRESU

QUALIFIERS

Everything else describing it

VSBLFL · VSPOS · LBNRIND · LBCAT

TIMING

When it was measured

VISITNUM · VISIT · VSDTC · VSDY

The two highlighted rows are what's new. In AE the topic was WHAT HAPPENED and the answer was the same variable. In Findings the topic is the QUESTION and the result is a SEPARATE set of variables.

Raw data is WIDE. SDTM is TALL.

RAW — vs_raw.csv • one row per VISIT, one column per measurement

SUBJID	VISIT	SYSBP	DIABP	PULSE	TEMP	HEIGHT	WEIGHT
001	SCREENING	122	80	68	36.7	165	70.5
001	BASELINE	120	78	66	36.8		70.2

1 subject × 2 visits = 2 rows

Add a new test → add a new COLUMN.
The structure changes every time the protocol does.



SDTM — VS • one row per SUBJECT × VISIT × TEST

USUBJID	VISIT	VSTESTCD	VSORRES	VSORRESU
ABC-01-01-001	SCREENING	SYSBP	122	mmHg
ABC-01-01-001	SCREENING	DIABP	80	mmHg
ABC-01-01-001	SCREENING	PULSE	68	beats/min
ABC-01-01-001	SCREENING	...	...	...

1 subject × 2 visits × 6 tests = up to 12 rows

Add a new test → add new ROWS.
The STRUCTURE never changes. That is the point.

Why bother? A reviewer can write ONE program that reads every Findings domain in every study — because they all have the same columns. A wide dataset needs a new program for every protocol.

DOING IT

Six output rows from one input row

The array-and-loop method — one OUTPUT per test, skipping what wasn't measured

SAS · wide → tall

```
array vals{6}  sysbp diabp pulse temp height weight;
array cds {6} $8 ('SYSBP' 'DIABP' 'PULSE' 'TEMP' 'HEIGHT' 'WEIGHT');

do i = 1 to 6;
    if not missing(vals{i}) then do;          /* no result -> no row */
        vstestcd = cds{i};
        vsorres  = put(vals{i}, best.);
        output;                               /* <-- one row per test */
    end;
end;
```

The three moving parts

1. an array of the VALUE columns
2. a parallel array of TEST CODES
3. OUTPUT inside the loop, not after it

⚠ The classic bug

OUTPUT placed AFTER the loop instead of inside it.
You get one row per visit carrying only the LAST test — and it looks plausible enough to ship.

Why SDTM stores the same number three times

--ORRES

CHARACTER

ORIGINAL result, exactly as collected

Never modified. This is the audit trail back to the CRF.

"120"

--STRESC

CHARACTER

STANDARDISED result, as character

Converted to the study's standard unit. Holds results that are NOT numbers, e.g. "POSITIVE".

"120"

--STRESN

NUMERIC

STANDARDISED result, as a number

The only one you can average. Empty when the result isn't numeric.

120

In ABC-01 all three look identical — because the sites already collect in standard units. That is a property of THIS study, not a rule. A site reporting weight in pounds would give ORRES 155 lb and STRESN 70.3 kg.

--BLFL: the variable every analysis leans on

What it means

“This is the LAST measurement taken before the subject was first treated.”

Almost every efficacy analysis is a CHANGE FROM BASELINE. If the flag is wrong, every treatment effect in the study is computed against the wrong starting point.

Values

“Y” this record is the baseline

null it is not

⚠ There is no “N”. The flag is Y-or-blank, and a stray “N” is a conformance finding.

In ABC-01 — the flag lands on BASELINE, never on SCREENING

VISIT	VSDTC	VSDY	SYSBP	VSBLFL	why
SCREENING	2024-02-20	-10	122	(null)	before, but not the last
BASELINE	2024-03-01	1	120	Y	last before first dose
WEEK 4	2024-03-28	28	118	(null)	after treatment started

Both SCREENING and BASELINE precede first dose — but only ONE record can carry the flag. It is the LAST one before treatment, not simply any earlier one.

40 of the 128 VS records carry VSBLFL = “Y” — 8 subjects × 5 tests at BASELINE (height isn't repeated).

Two ways of saying when — and why you need both

VISIT

The visit NAME, as the protocol calls it.

"SCREENING" · "BASELINE" · "WEEK 4"

Readable, but it sorts ALPHABETICALLY — which puts WEEK 4 before other weeks and BASELINE first by accident.

VISITNUM

The visit's NUMBER in protocol order.

1 · 2 · 4

Numeric, so it sorts CHRONOLOGICALLY. This is what every analysis actually orders and groups by.

The mapping is a lookup from the protocol — it is not derivable from the data

VISIT	VISITNUM	note
SCREENING	1	up to 28 days before first dose
BASELINE	2	day of first dose — carries VSBLFL
WEEK 4	4	VISITNUM 3 is not used in VS or LB

VISITNUM 3 is deliberately absent. Gaps are normal — the numbers come from the protocol's visit schedule, which may include visits where no vital signs or labs are taken. Never renumber them to close a gap.

LB is VS plus reference ranges

LB raw arrives ALREADY TALL — one row per test — so there is no transpose. What it adds instead is the machinery for deciding whether a result is normal.

LBCAT	The panel the test belongs to	HEMATOLOGY · CHEMISTRY
LBORNRL0 / LBORNRHI	Reference range low and high, in the ORIGINAL units	12.0 ... 17.0
LBSTNRLO / LBSTNRHI	The same range in STANDARD units	12.0 ... 17.0
LBNRIND	Where the result falls: derived, not collected	NORMAL · LOW · HIGH
Deriving LBNRIND <pre>if lbstresn < lbstnrlo then lbnrind = "LOW"; else if lbstresn > lbstnrhi then lbnrind = "HIGH"; else lbnrind = "NORMAL";</pre>		

47 normal results, and one that isn't

USUBJID	VISIT	TESTCD	ORRES	NRLO	NRHI	NRIND	LBCAT	LBDY
ABC-01-01-003	BASELINE	ALT	24	7	56	NORMAL	CHEMISTRY	1
ABC-01-01-003	WEEK 4	ALT	72	7	56	HIGH	CHEMISTRY	28

What the data says

Subject ABC-01-01-003's ALT (a liver enzyme) was 24 at Baseline — comfortably normal — and 72 at Week 4, above the upper limit of 56.

LBNRIND is HIGH because the result exceeds LBSTNRHI. That is arithmetic, not judgement.

What the data does NOT say

That the subject was harmed. “HIGH” is a range comparison, nothing more.

Whether an abnormal lab is CLINICALLY SIGNIFICANT is a physician's judgement, recorded separately — and if it is, it becomes an adverse event in AE.

The programmer's line

You derive LBNRIND. You do NOT decide clinical significance, and you never create an adverse event because a lab value looks bad. That subject has no ALT-related AE in ABC-01 — and inventing one to “make the data consistent” would be fabricating a safety finding.

The lab test name you were sent is not the one you submit

RAW LBTEST (from the lab)	LBTESTCD	SDTM LBTEST (CDISC CT)	same?
Hemoglobin	HGB	Hemoglobin	yes
Hematocrit	HCT	Hematocrit	yes
White Blood Cells	WBC	Leukocytes	NO — renamed
Platelets	PLAT	Platelets	yes
Alanine Aminotransferase	ALT	Alanine Aminotransferase	yes
Creatinine	CREAT	Creatinine	yes

“White Blood Cells” is what the lab calls it. “Leukocytes” is what CDISC calls it. Five of six pass straight through, so a programmer who spot-checks two rows concludes the column maps 1-to-1 — and ships the sixth wrong.

Both columns are controlled terminology — check both

LBTESTCD is the short code (max 8 characters, no spaces) and LBTEST is the full name. Both come from the CDISC LBTESTCD/LBTEST codelists, and they are PAIRED — you cannot mix a code from the list with a name of your own. Look every test up; never translate by eye.

Same class, different challenges

	VS • Notebook 08	LB • Notebook 09
Raw shape	WIDE — 1 row per visit, 6 value columns	TALL — already 1 row per test
Main challenge	the transpose	reference ranges and CT lookups
Raw rows	24	48
SDTM rows	128	48
Tests	6 (height only at screening)	6 across 2 panels
Visits	SCREENING, BASELINE, WEEK 4	BASELINE, WEEK 4
Baseline flags	40	24
Derived qualifier	VSBLFL	LBBLFL, LBCAT, LBNRIND
Watch out for	OUTPUT inside the loop; height's missing values	“White Blood Cells” → Leukocytes

Note the row counts: VS goes from 24 raw rows to 128 SDTM rows, while LB stays at 48. That difference IS the transpose.

The five that cost the most time

1

OUTPUT after the loop

You get 1 row per visit instead of 6. The dataset looks fine until you count it.

→ count your rows first, always

2

Arrays out of order

The value array and the test-code array must line up exactly, or every pulse is labelled a temperature.

→ read the two ARRAY statements side by side

3

Rows for values that weren't measured

Height is collected once. Emitting a blank HEIGHT row at every visit inflates the domain with nothing.

→ test NOT MISSING before OUTPUT

4

Flagging baseline on SCREENING

--BLFL is the LAST record before first dose, not the first record in the study.

→ check it lands on exactly one record per subject per test

5

Translating test names by eye

Five of six lab tests map straight through. The sixth does not.

→ look up every value in the codelist